## Q1
2 Points

You just took a medical test to see if you contracted a disease. The test comes back positive. Your doctor tells you that the test is very good: apparently it has a Sensitivity of 99%. How convinced should you be that you have the disease? Options:

A. Very. You definitely have the disease

B. It depends. More information about the test is required.

C. You definitely do not have the disease.

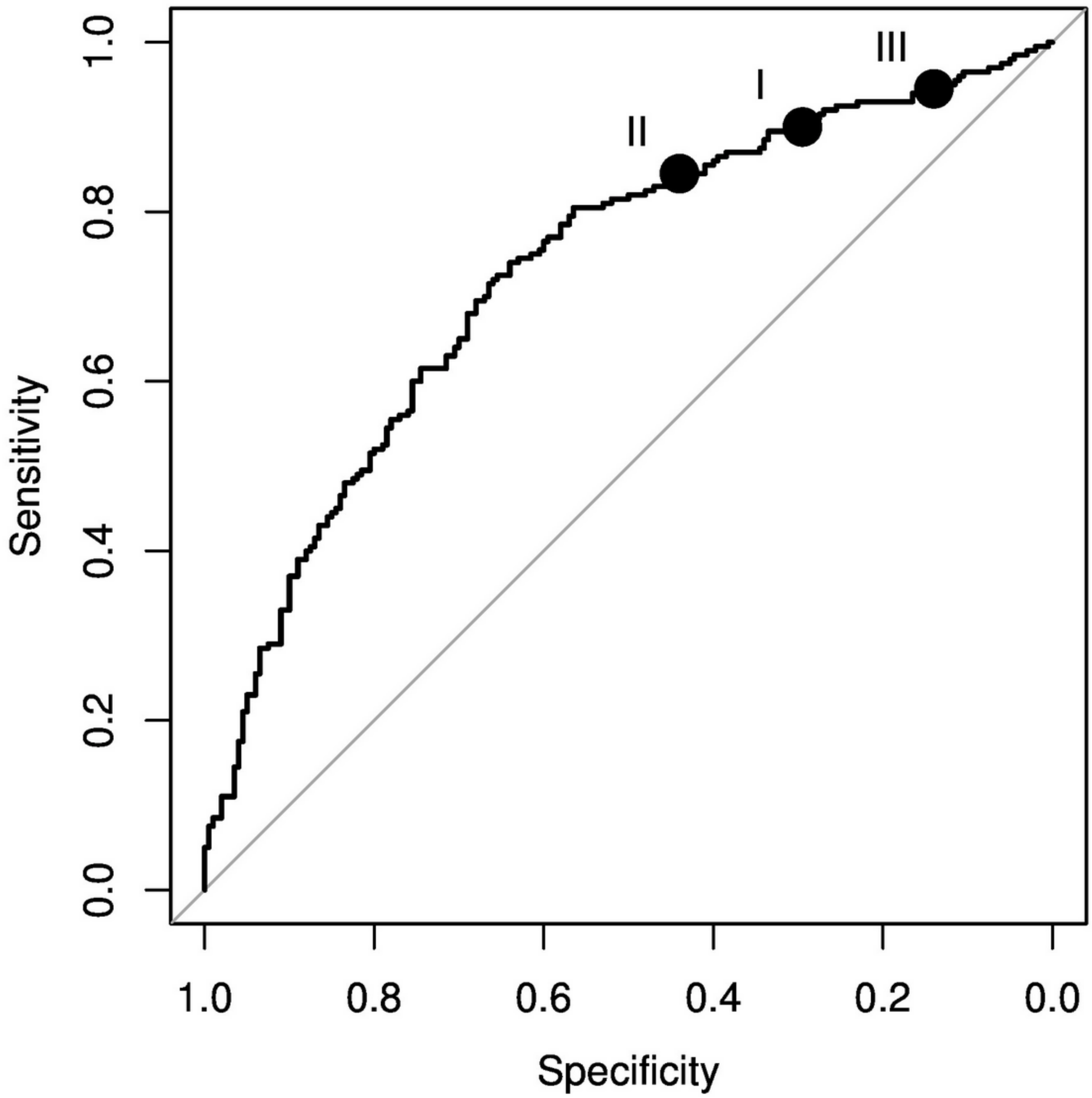
Choose an answer and **explain your reasoning**:

If you answered B, what metric of the test would you want to know instead?

Sensitivity captures $P(\text{guess } 1|Y=1)$. However, this doesn't say anything about whether you have it. A test that always guessed 1 would have perfect sensitivity. You want $P(Y=1|\text{guess } 1)$, which is PPV (aka precision).

## Q2

2 Points



I
II
III
Sensitivity
0.0
0.2
0.4
0.6
0.8
1.0
Specificity
1.0
0.8
0.6
0.4
0.2
0.0

Suppose that the ROC curve above reflects a classifier that you use for identifying marketing targets. For years, you've been using a threshold that corresponds to point I on the plot. New management suddenly decides that false leads (false positives) are more expensive to follow up on than you thought (shifting $L_{FP}/L_{FN}$ upward)! Should you move your threshold to give a point in the direction of point II or point III?

- (selected) point II
- ( ) point III

How should you change your threshold on your estimated $P(Y = 1|X = x)$ to achieve this shift?

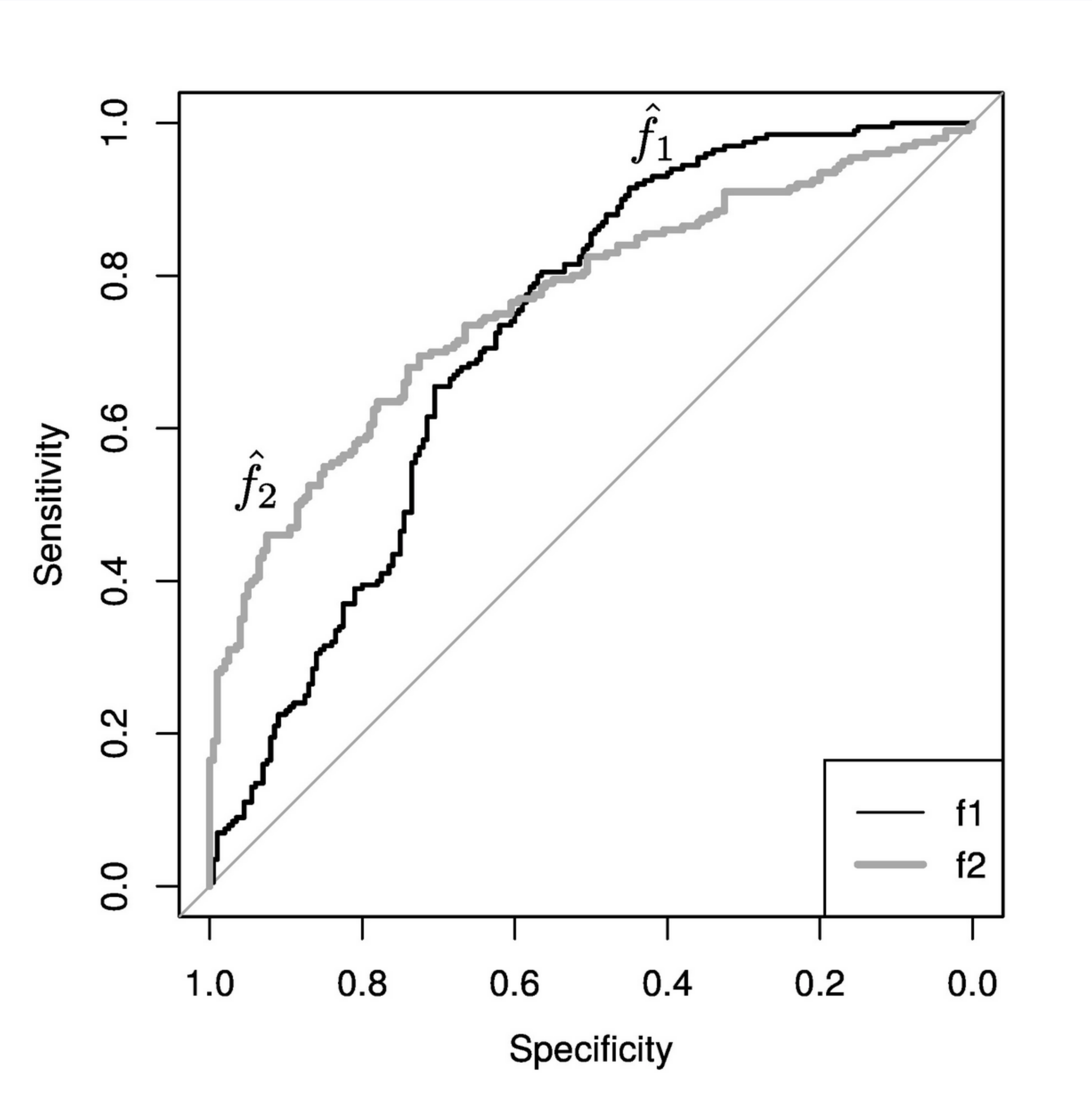
- (selected) Make the threshold higher.
- ( ) Make the threshold lower.

## Q3

1 Point

Suppose that you are building a model for fraud detection. Fradulent transactions are very expensive for you to miss, but it is not terribly expensive to follow up on a reasonable number of non-fradulent transactions.

You construct two classification scores, $\hat{f}_1(x)$ and $\hat{f}_2(x)$. The ROC curves for each classification score are shown below.




Based on these ROC curves and what you know of your fraud detection scenario, which classification score would you choose? (Assume that you would then choose an appropriate threshold to obtain your actual classifier.)

- (selected) I would choose $\hat{f}_1(x)$.
- I would choose $\hat{f}_2(x)$.

## Q4

1 Point

Suppose that we have written a classifier to identify clients who will likely be profitable ( $Y = 1$ indicating profitable, $Y = 0$ indicating unprofitable). Our model $\hat{f}(x)$ outputs values between 0 and 1, with high values indicating individuals that are more likely to be profitable clients. At a score cutoff of $s = 0.5$, we find that the sensitivity of the classifier is too low. How can we try to improve the Sensitivity of our classifier?

- (•) Try a value of $s$ that is smaller than 0.5.
- ( ) Try a value of $s$ that is greater than 0.5.
- ( ) The only way to improve sensitivity is to build a different model.

## Q5

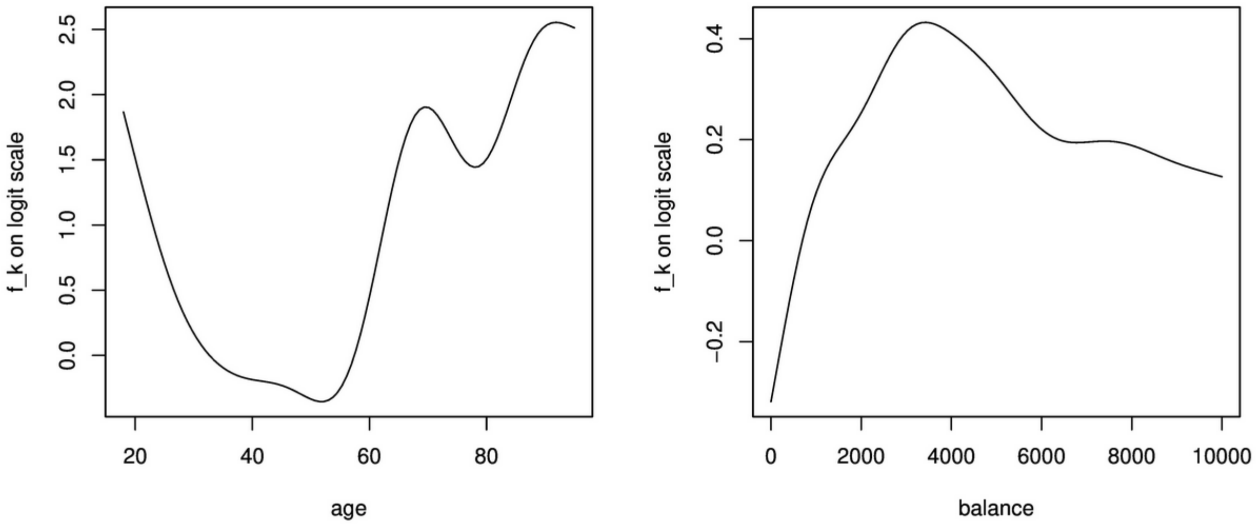
2 Points

Suppose that you fit a logistic generalized additive model to the marketing data that you saw in your homework. Your model predicts success in marketing as a function only of **age** and **balance**. You fit the model

$$\log\left(\frac{P(\text{Success}|X = x)}{1 - P(\text{Success}|X = x)}\right) = \beta_0 + \hat{f}_{age}(\text{age}) + \hat{f}_{balance}(\text{balance}).$$

This is exactly like logistic regression, except that your covariates enter through smooth functions instead of linear combinations (just like additive models).

The corresponding fits $\hat{f}_{age}$ and $\hat{f}_{balance}$ of those smooth functions are shown below. Note that we are modeling the logit probabilities on the left side, so each function's values should be interpreted on that logit scale. **The intercept is estimated to be $\hat{\beta}_0 = -2.02$.**




Suppose that you use the usual 0-1 loss threshold of $P(Y = 1|X = x) > 0.5$ for your classifier. Explain why your classifier will never predict success for a person between 30 and 60 years old.

For individuals with ages between 30 and 60, the contribution from the age function is $< 0.5$. The other function is never above about 0.4, and the intercept is -2.02. Therefore the logit predictions are always $< -1.1$, so never cross zero.

Given that your classifier will never classify a person between 30 and 60 years old as a success, what value might your model have to marketing efforts in this age range?

(Just for your comfort, this problem has no relationship to or dependence on the ungraded generalized additive model problem on the homework. Just using basic ideas from class.)

Though the classifier would never predict a 1 when thresholding at 0.5, the probabilities can still be used to prioritize more promising marketing targets within this segment. (Also accept pretty much anything reasonable about asymmetric losses, different thresholds, etc.)